GRAVEATLAS

Phase 5 — Advanced Search, Discovery, Personalization & User Experience

Full copy-paste implementation prompt for the Base44 agent.

COPY-PASTE MASTER PROMPT

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GRAVEATLAS — PHASE 5
ADVANCED SEARCH, DISCOVERY, PERSONALIZATION & USER EXPERIENCE

Phase 1, Phase 2, Phase 3 and Phase 4 must already be complete.

Continue from the existing GraveAtlas implementation.

IMPORTANT:
- Do NOT rebuild Phases 1–4.
- Reuse the existing Android app, Cloudflare Worker/API, GitHub App, public data repository, schemas, search, map, authenti
- Do NOT start Phase 6.
- Do NOT add paid services without explicit approval.
- Do NOT introduce unnecessary external AI services.
- Do NOT fabricate cemetery, grave, person or historical information.
- Public facts must remain grounded in the approved dataset and provenance.
- Do NOT expose private contributor, moderator or administrator information.
- Do NOT weaken existing security or authorization.
- Complete only Phase 5, then stop.

==================================================
PART 1 — SEARCH ARCHITECTURE REVIEW
==================================================

Review the Phase 2 search implementation.

Preserve the existing public API contract where practical.

Identify opportunities to improve:

- relevance
- response speed
- filtering
- sorting
- typo tolerance
- normalized search
- geographic discovery

Do not replace working infrastructure without a clear benefit.

==================================================
PART 2 — ADVANCED CEMETERY SEARCH
==================================================

Enhance cemetery discovery.

Support appropriate filters such as:

- cemetery name
- city/region
- country
- geographic area
- available record count
- data/source status where public

Only expose fields that are genuinely available.

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PART 3 — ADVANCED PERSON/RECORD SEARCH
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Enhance record search.

Support appropriate filters such as:

- name
- cemetery
- date
- location
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- record type

Do not imply that two similarly named people are the same person without evidence.

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PART 4 — SEARCH RELEVANCE
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Implement a deterministic relevance strategy.

Possible factors:

- exact match
- prefix match
- normalized match
- token match
- cemetery match
- geographic relevance

Document the actual scoring approach.

Do not use an opaque external AI service merely to rank simple public records.

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PART 5 — TYPO / VARIANT HANDLING
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Where practical support common search variations.

Handle:

- punctuation
- spacing
- capitalization
- Unicode variants
- common transliteration differences

Do not silently alter authoritative records.

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PART 6 — SEARCH FILTERS
==

Implement usable filters without making the search interface unnecessarily complex.

Filters must be:

- validated
- bounded
- reflected in API requests
- preserved when navigating results

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PART 7 — SEARCH SORTING
==

Support appropriate sorting options where useful.

Examples:

- relevance
- alphabetical
- distance where location is available
- recently updated public data

Do not present an unsupported "verified" or "official" sort.

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PART 8 — SEARCH RESULT QUALITY
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Improve result cards.

Show enough information to distinguish records without exposing private data.

Avoid misleading truncated names or dates.

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PART 9 — SEARCH HISTORY
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If local search history is implemented:

- keep it local where practical
- allow clearing
- do not store unnecessary sensitive information
- do not send history to the server without a valid reason

If search history is not needed, do not add it merely for feature count.

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PART 10 — RECENTLY VIEWED
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If appropriate, provide locally stored recently viewed public records/cemeteries.

Do not store private moderation information.

Allow clearing local history.

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PART 11 — FAVORITES / BOOKMARKS
==

If appropriate, allow users to bookmark public cemetery/record pages.

Prefer local storage unless an authenticated synchronized system already exists.

Do not turn bookmarks into public data.

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PART 12 — MAP DISCOVERY
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Enhance map discovery using the existing map foundation.

Support appropriate:

- viewport-based search
- marker clustering
- cemetery selection
- result-to-map navigation
- map-to-result navigation

Keep geographic queries bounded.

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PART 13 — DISTANCE-BASED DISCOVERY
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If the user grants location permission:

allow optional nearby discovery.

Important:

- location permission must not be required for normal browsing
- do not collect location unnecessarily
- do not store precise user location unless explicitly required
- clearly explain permission use

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PART 14 — LOCATION PRIVACY
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Verify that precise user location is not accidentally sent to the backend.

If nearby search requires server-side coordinates:

use the minimum information necessary.

Do not retain user location unnecessarily.

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PART 15 — CEMETERY DISCOVERY
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Create a discovery experience for browsing cemeteries.

Possible sections:

- nearby
- recently viewed
- popular only if real usage data exists
- alphabetical
- region
- map

Do not fabricate popularity metrics.

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PART 16 — RECORD DISCOVERY
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Provide useful pathways from:

cemetery
→ sections/records
→ person/record
→ source
→ location

Do not create relationships that are not supported by the data.

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PART 17 — RELATED RECORDS
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Where relationships are explicitly present in the dataset, show related records.

Examples:

- same cemetery
- same section
- same family/relationship if reliably sourced
- nearby records

Do not infer family relationships from matching surnames alone.

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PART 18 — SOURCE TRANSPARENCY
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Make source/provenance information easier for users to understand.

Clearly distinguish:

- source
- contribution
- moderation status where publicly appropriate
- publication date/version

Do not expose private moderation notes.

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PART 19 — DATA CONFIDENCE LANGUAGE
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If the system uses confidence or quality indicators:

ensure terminology does not imply historical certainty beyond the evidence.

Use clear labels such as:

SOURCE AVAILABLE
MULTIPLE SOURCES
NEEDS REVIEW

only when those states actually exist.

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PART 20 — USER EXPERIENCE
==

Review the complete discovery journey:

HOME
→ SEARCH
→ RESULTS
→ CEMETERY
→ RECORD

→ MAP
→ SOURCE

Reduce unnecessary navigation steps.

Do not redesign working screens without a usability reason.

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PART 21 — HOME SCREEN
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Improve the home screen so the main public actions are immediately understandable.

Prioritize:

- search
- browse cemeteries
- map
- contributions where applicable

Do not overload the home screen with unrelated features.

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PART 22 — EMPTY STATES
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Improve empty states for:

- no search results
- no nearby cemeteries
- no bookmarks
- no history
- no records
- no map results

Each state should offer a useful next action.

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PART 23 — ERROR STATES
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Improve errors for:

- network unavailable
- API unavailable
- timeout
- invalid search
- map unavailable
- stale data

Provide safe retry actions.

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PART 24 — OFFLINE EXPERIENCE
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Review the existing offline foundation.

Where practical, allow users to continue viewing previously cached public information.

Clearly indicate stale/offline data.

Do not claim data is current when it is cached.

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PART 25 — PERFORMANCE
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Optimize:

- search latency
- map loading
- result rendering
- pagination
- cache usage
- Android memory usage

Avoid downloading unnecessary datasets.

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PART 26 — ACCESSIBILITY
==

Review Phase 5 screens for:

- content descriptions
- readable text
- touch targets
- screen-reader labels
- scalable text
- keyboard/navigation support where applicable

Do not introduce accessibility regressions.

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PART 27 — ANALYTICS PRIVACY
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If analytics exist:

review collection.

Do not collect:

- unnecessary precise location
- private contribution content
- moderation notes
- secrets
- sensitive personal information

If analytics do not exist, do not add a paid analytics service.

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PART 28 — PERFORMANCE OBSERVABILITY
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Where existing monitoring supports it, track safe operational metrics such as:

- API response time
- search failures
- map failures
- error rates

Do not log personal search history unnecessarily.

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PART 29 — API OPTIMIZATION
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Review public API queries.

Ensure:

- bounded results
- efficient filters
- pagination
- caching
- safe query parameters

Avoid unbounded database or repository scans.

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PART 30 — SECURITY
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Verify Phase 5 features do not create:

- unauthorized data access
- IDOR vulnerabilities
- location leakage
- excessive API access
- search abuse
- private-data exposure

All authorization remains server-side.

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PART 31 — TESTING
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Test:

- advanced cemetery search
- advanced record search
- filters
- sorting
- typo handling
- map discovery
- nearby discovery
- location permission denial
- bookmarks/history if implemented
- offline browsing
- caching
- accessibility
- API limits
- unauthorized access

Do not fabricate results.

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PART 32 — DOCUMENTATION
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Create/update:

docs/ADVANCED-SEARCH.md
docs/DISCOVERY.md
docs/MAP-DISCOVERY.md
docs/LOCATION-PRIVACY.md
docs/OFFLINE-MODE.md
docs/USER-EXPERIENCE.md
docs/SEARCH-PERFORMANCE.md

Document actual implementation only.

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PHASE 5 ACCEPTANCE GATE
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Verify:

[ ] Advanced cemetery search
[ ] Advanced record search
[ ] Search relevance
[ ] Typo/variant handling
[ ] Filters
[ ] Sorting
[ ] Result quality
[ ] Search history if implemented
[ ] Recently viewed if implemented
[ ] Bookmarks if implemented
[ ] Map discovery
[ ] Nearby discovery
[ ] Location privacy
[ ] Cemetery discovery
[ ] Record discovery
[ ] Related records
[ ] Source transparency
[ ] Confidence/quality language
[ ] User experience
[ ] Home screen
[ ] Empty states
[ ] Error states
[ ] Offline experience
[ ] Performance
[ ] Accessibility
[ ] Analytics privacy
[ ] Observability
[ ] API optimization
[ ] Security
[ ] Tests
[ ] Documentation

==
FINAL REPORT
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Return:

PHASE 5 STATUS:

READY / NOT READY

SEARCH:
PASS / FAIL

DISCOVERY:
PASS / FAIL

MAP:
PASS / FAIL

LOCATION PRIVACY:
PASS / FAIL

USER EXPERIENCE:
PASS / FAIL

OFFLINE:
PASS / FAIL

PERFORMANCE:
PASS / FAIL

ACCESSIBILITY:
PASS / FAIL

SECURITY:
PASS / FAIL

API:
PASS / FAIL

TESTS:
PASS / FAIL

DOCUMENTATION:
PASS / FAIL

==
ACTUAL TEST RESULTS
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Report actual results only.

Cemetery search:
[actual result]

Record search:
[actual result]

Filters:
[actual result]

Map:
[actual result]

Nearby:
[actual result]

Offline:
[actual result]

Accessibility:
[actual result]

Performance:
[actual result]

Security:
[actual result]

Do NOT invent results.

==
BLOCKERS
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List every issue preventing:

PHASE 5 = READY

For each:
- severity
- component
- problem
- impact
- recommended remediation

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NON-BLOCKING ISSUES
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List remaining improvements.

==
STOP CONDITION
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Do NOT start Phase 6.

Do NOT add paid services without explicit approval.

Do NOT add unnecessary external AI services.

Do NOT collect unnecessary location or personal data.

Do NOT fabricate data or test results.

Do NOT expose secrets.

Do NOT weaken authorization.

Do NOT delete legitimate cemetery or historical data.

STOP after the final report.

WAIT FOR EXPLICIT INSTRUCTIONS FOR PHASE 6.